

Nathan Stefanik

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EXPERIENCE

JetBlue Airways

Jun 2023 – Present

Data Scientist

New York, NY

- Built and maintained a production-oriented GLMM pipeline that generates Core and Mint spill-spoil curves for Revenue Management (RM) review – Python/Polars feature stages, R glmmTMB training, Snowflake inputs, Databricks jobs, and MLflow model handling.
- Moved MLflow run lifecycle, artifact logging, and registry operations to Python while retaining R for fit/score I/O, producing packageable Databricks model artifacts.
- Defined experiment-comparison metrics on prediction bands (denominator-weighted in-band rate, coverage, bandwidth, tail imbalance) and improved a Tableau validation workbook so coverage alone is not treated as quality.
- Designed a slice-first red-flag framework on optimized-price deltas (grain health, directional correctness, false positives, ranked slices) and a 30-minute decision-oriented MVP.
- Propagated exact MLflow model IDs through prediction, post-processing, evaluation, and production-candidate tables so evaluation can distinguish models rather than only groups.
- Identified a shared-SQL mismatch that mixed cabin seat-factor data and implemented task-owned cabin parameters for Core (Y) and Mint (J).

Invesco Ltd

Jun 2022 – Aug 2022

Info Risk and Controls Intern

Atlanta, GA

- Provided analysis to Global Risk and Controls, including an incident-ticket classification model (Doc2Vec + GMM) at 70% accuracy, and processed large datasets from Microsoft SQL Server in Python (pyodbc).
- Built PowerApps for internal risk management and control validation.

Georgia Tech School of Mathematics

Fall 2021

Grader, Linear Algebra II

Atlanta, GA

- Graded problem sets for over 70 students and created answer keys and rubrics.

EDUCATION

Georgia Institute of Technology

Dec 2022

BS, Mathematics and Computer Science; GPA 3.6/4; Highest Honors

Atlanta, GA

- Concentration in Machine Learning and Enterprise Computing. Zell Miller Scholarship, 2018–2022.
- Math – Real Algebraic Geometry and Optimization, Algebraic Topology, Statistical Theory, Probability Theory, Real Analysis, Complex Analysis, Abstract Algebra.
- CS – Database System Implementations, Deep Learning, Machine Learning, Honors Algorithms, Intro to AI, Computer Systems and Networks.

PROJECTS

IBKR paper-trading infrastructure | Python, Polars, DuckDB

- Paper-only Interactive Brokers stack (ib_async) with a kill switch (1% daily loss cap, 5% equity-path position cap), paper fills, Docker/systemd, and a vectorized weight-space backtest with EWMA vol on a 12-instrument CME-group universe. Not live trading.

BTS fare intelligence | Polars, public DOT data

- Independent Polars pipelines over public DB1B (10% ticket sample) and T-100 traffic. Pre-MVP; exploratory ingest and cleaning, not a shipped consumer product.

Lean 4 practice and chi-squared | Lean 4, Mathlib

- Practice proofs (calc, rw, simp, ring, linarith, induction) and an in-progress chi-squared formalization as gamma with shape $p/2$. No Mathlib PR.

WRITING

The t-statistic is t-distributed

2026

nathanstefanik.xyz/blog/math_t_statistic.html

- Proved that the OLS coefficient t-statistic follows Student-t with $n-p$ degrees of freedom under Gaussian errors. Not peer-reviewed.

daily.nathanstefanik.xyz

https://daily.nathanstefanik.xyz

- Daily quant-interview and ML/stats problems, generated in batches and published statically.

Course and DRP notes

2021–2022

nathanstefanik.xyz/projects/

- Notes on the Conway knot, algebraic cycles, abelian varieties, characteristic classes, tropical polynomials, playlist danceability, and Fashion-MNIST in Julia/Flux.

TECHNICAL SKILLS

Languages and data: Python, SQL, Snowflake, Polars, DuckDB, R

Statistics and models: GLMM (glmmTMB), statistical prediction intervals, experiment comparison (coverage, bandwidth, in-band rate)

Platform: Databricks, MLflow, Tableau, Docker

Domain: airline revenue management, continuous pricing, demand forecasting